(Q1) You are given the demand distribution for one week

| Demand | Probability |
|---|---|
| 0 | 0.2 |
| 1 | 0.5 |
| 2 | 0.2 |
| 3 | 0.1 |

What is the demand distribution for four weeks.

(1) The Expected Value $E(x)$ & Variance $Var[x]$ for one week demand $X$ are

$E(X) = \sum x_i\, p(x_i)$

$= 0 \times 0.2 + 1 \times 0.5 + 2 \times 0.2 + 3 \times 0.1$

$= 1.2$

$E(x^2) = \sum x_i^2\, p(x_i) = 0^2 \times 0.2 + 1^2 \times 0.5 + 2^2 \times 0.2 + 3^2 \times 0.1 = \underline{2.3}$

$$Var(x) = E(x^2) - (E(x))^2$$
$$= 2.2 - (1.2)^2 = 0.76$$

(2) Calculate mean & Variance for four weeks

For a four week period $y = x_1 + x_2 + x_3 + x_4$

Mean & Variance are

$$E(y) = 4 \times E(x) = 4 \times 1.2 = 4.8$$
$$Var.(y) = 4 \times Var(x) = 4 \times 0.76 = 3.04$$

(3) Determine the full distribution (using Convolution)

The possible demand values range from
$0 \times 4 = 0$ to $3 \times 4 = 12$

Let $P_1$ - Single week probability mass function (PMF)

$P_4$ - four week PMF

$$P_4 = P_1 \times P_1 \times P_1 \times P_1$$

Q.2 Weekly demand for an item is normally distributed with mean equal to 100 & S.D = 8, Ordering cost is $10 & carrying cost is 12% per yr. The price per item is $12. There is a two-week lag b/w order and delivery. Assume there is a fixed under stock cost of $200. Using the approximate method of analysis, find the optimal Q-system for this item.

(b) What is the total cost for this system

(c) Use the exact method of analysis to find the optimal Q-system

Identify parameters

$\mu$ – weekly demand = 160 units

$\sigma$ – weekly SD = 8 unit

A – ordering cost = \$10

$C_h$ – Carrying cost per year

= 12% of price

= 0.12 × \$12

= \$1.44 per unit per year

$C_u$ = Understock (shortage) cost

= \$200 per unit

Lead time, L = 2 weeks

price per item = \$12

(1) Calculate annual demand & Optimal order quantity (EOQ)

Annual demand, $D = 52 \times \mu$

$= 52 \times 160$

$= 5200$ units

Economic Order Quantity (EOQ) is $Q^*$

$$Q^* = \sqrt{\frac{2AD}{C_h}} = \sqrt{\frac{2 \times 10 \times 5200}{1.44}}$$

$$= \underline{\underline{268.74 \text{ units}}}$$

(2) Determine Optimal Reorder point (R) using approximate analysis.

Demand during lead time $\mu_L = L \times \mu$

$= 2 \times 100$

$= 200$ units

Standard D. during lead time, $\sigma_L = \sigma \times \sqrt{L}$

$= 8 \times \sqrt{2}$

$= \underline{\underline{11.312}}$

The optimal service level P (probability of no-stock per cycle)

$$P = 1 - \frac{A}{C_u D} \text{ or } 1 - \frac{C_h Q}{C_u D}$$

$$P = 1 - \frac{1.44 \times 268.74}{200 \times 5200} = \underline{\underline{0.9996}}$$

The reorder point R is

$\mu_L + Z\sigma_L$

Z – safety factor corresponding to p

i.e from standard normal tables

p of 0.9996 = 3.35 ~ Z

SS – safety stock

$= Z\sigma_L = 3.35 \times 11.312$

$= 37.935$ units

$R \simeq \mu_L + Z\sigma_L$

$= 200 + 37.935 = 237.935$ units

(3) Total cost for this item (TC)

$TC = \frac{D}{Q}A + \frac{Q}{2}C_h + S.S \times C_h +$

Expected shortage cost

Expected shortage cost is very low.

$$TC = \frac{5200}{268.74} \times 60 + \frac{268.74}{2} \times 1.44 +$$

$$37.935 \times 1.44 + \approx 2150.$$

$$TC = \$\underline{\underline{441.60}}$$

(4) Use the exact method of analysis to find the optimal Q-system.

The exact method involves finding the optimal Q and R simultaneously by minimizing the total cost function, often using iterative techniques or specific formulas involving the unit normal loss function E(z). The optimal Q is found from the ratio $\frac{C_h Q}{C_u D}$

The process is iterative because Q depends on Z (via safety stock) & Z depends on

Q ≃ 269 units

R = 238 units

cost = \$ 441.60

Q3 For the item of problem (2)
use the approximate method of analysis to find the optimal P-system for the item. Make the assumption that the reserve stock is carried continuously in calculating carrying costs.
(b) What is the total cost for this system
(c) Use the exact method of analysis to find the optimal P-system for this item.

(1) Identify the parameters & calculate the Optimal Review period (T)

Parameter are same as pblm. (2).

The optimal Review period $T^*$ is (during 52 weeks)

$$T^* = \frac{Q^*}{D} = \frac{268.74}{5200}$$

$$= 2.687 \text{ Weeks}$$

$$\simeq 2.7 \text{ Weeks}$$

(2) Determine optimal order-up-to level (M) using approximate analysis

The order-up-to level M is based on demand during the protection interval.

$$T+L = 2.687 + 2 = 4.687 \text{ week}$$

Mean demand during protection interval

$$\mu_{T+L} = \mu \times (T+L)$$
$$= 100 \times 4.687$$
$$= 468.7 \text{ units}$$

Standard deviation during protection interval

$$\sigma_{T+L} = \sigma \times \sqrt{T+L}$$
$$= 8 \times \sqrt{4.687}$$
$$= 17.32 \text{ units}$$

Using the same service level $p \simeq 0.9996$

$z\ (3.35)$

$\therefore$ Safety stock $SS = z \times \sigma_{T+L}$

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$= 3.35 \times 17.32 = 58.02$ units

$M = \mu_{T+L} + SS$

$= 526.72$ units

(3) Total cost for this system

$$TC = \frac{D}{T} A + C_h \times (\text{Avg. Inventory})$$

$$\text{Avg Inventory} = \frac{DT}{2} + SS$$

$$TC = \frac{5200}{0.05168} \times 10 + C_h \times \left( \frac{5200 \times 0.05168}{2} + 58.02 \right)$$

$= \$1{,}066.477$

(4) $T \simeq 2.7$ weeks

$M = 527$ units. $TC = \$1{,}066.9$